

Economic Growth, the Financial System, and Business Cycles

ECON 101 - Macroeconomics

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Fall 2026

- **6.1 Long-Run Economic Growth**
- 6.2 Saving, Investment, and the Financial System
- 6.3 The Business Cycle

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6.1 Learning Objective

Goal

Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising **productivity** increases the **average standard of living**.
- This is different from short-run fluctuations (**business cycles**):
 - Expansion: real GDP rising.
 - Recession: real GDP falling.
- The most common measure of the standard of living is:
 - Real GDP per capita = real GDP divided by population.

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Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
 - How much goods and services, on average, each person can consume.
 - Adjusted for changes in the price level (**real**, not nominal).
- It does not capture everything about well-being, but:
 - Countries with higher real GDP per capita typically have better:
 - health, education, housing, and infrastructure.

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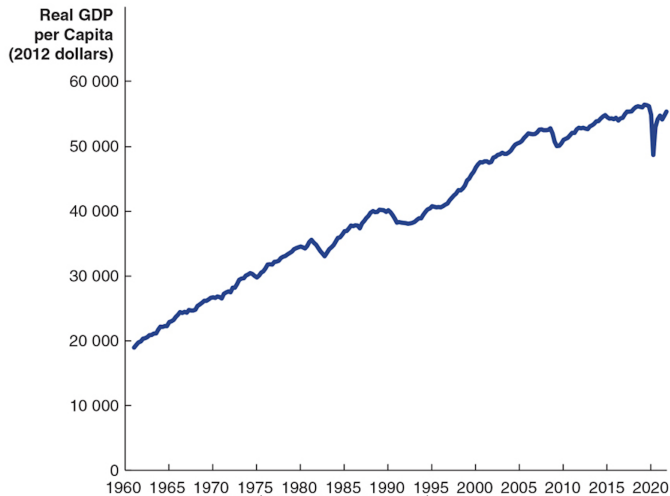
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Figure 6.1 – Real GDP per Capita, 1961–2021



Interpretation of Figure 6.1

- Measured in 2012 dollars, real GDP per capita in Canada has risen more than **three-fold** since 1961.
- Interpretation:
 - The average Canadian in 2021 can buy almost **three times as many** goods and services as the average Canadian in 1961.
- This long-run rise is what we mean by **economic growth**.
- Short-run recessions are bumps around this upward trend.

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Calculating Growth Rates

- Growth rate of an economic variable (e.g., real GDP) from year $t - 1$ to year t :

$$\text{Growth rate}_t = \frac{\text{Value}_t - \text{Value}_{t-1}}{\text{Value}_{t-1}} \times 100.$$

- For a few years, we can compute growth each year and then take the average.
- Example:
 - Real GDP per capita grows by 2%, 3%, and 1% over three years.
 - Average annual growth $\approx (2 + 3 + 1)/3 = 2\%$.

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Growth Rates over Longer Periods

- Over longer periods, we use **compound growth**.

- Suppose:

- Previous real GDP = Y_0 ,
- Current real GDP = Y_t ,
- Annual growth rate = g ,
- Number of years between = t .

- Then:

$$Y_0(1 + g)^t = Y_t.$$

- Solving for g gives the average annual growth rate over the period.

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The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years it takes to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- Example:
 - If real GDP per capita grows at 5% per year:
 - Years to double $\approx 70/5 = 14$ years.
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Rule of 70: Classroom Example

- If Country A grows at 2% per year:
 - It takes about $70/2 = 35$ years to double income per person.
- If Country B grows at 4% per year:
 - It takes about $70/4 = 17.5$ years to double.
- After a few decades, Country B will be much richer than Country A, even though the growth rate difference seems small.

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What Determines Long-Run Growth?

- Long-run increases in real GDP per capita depend on increases in **labour productivity**:

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Number of workers or hours worked}}.$$

- The average Canadian today consumes more because:
 - the average Canadian produces much more per hour than in 1961.
- So, what makes workers more productive?

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 - Machines, computers, tools, buildings, vehicles.
- The more capital each worker has, the more they can produce per hour.
- Also important: **human capital:**
 - The accumulated **knowledge and skills** workers possess (education, training, experience).
- Example:
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- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
 - New production processes,
 - Better management techniques,
 - New products (e.g., smartphones, advanced medical devices).
- With better technology, workers can produce more **with the same amount** of capital and labour.
- **Entrepreneurs** play a critical role:
 - They pioneer new ways to bring together labour, capital, and ideas.

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- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
 - "Normal" hours,
 - "Normal" sized workforce,
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- Potential GDP increases when:
 - labour force grows,
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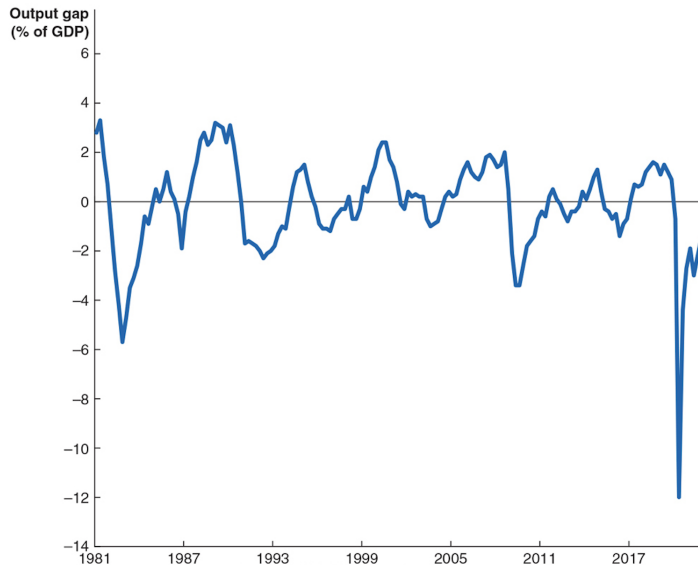
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Figure 6.2 – The Output Gap



The Output Gap

- **Output gap:** difference between actual GDP and potential GDP.
- When the output gap is **positive**:
 - actual GDP is above potential,
 - the economy is using resources in an **unsustainable** way (very low unemployment, high capacity usage).
- When the output gap is **negative**:
 - actual GDP is below potential,
 - resources (labour and capital) are **underutilized**.
- The Bank of Canada pays close attention to the output gap when setting interest rates.

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 - the economy is using resources in an **unsustainable** way (very low unemployment, high capacity usage).
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Goal

Explain how the financial system links saving and investment and why that matters for growth.

- Long-run growth requires:
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- Both require **investment**.
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An Overview of the Financial System

- **Financial markets:** markets where financial securities are bought and sold.
 - Examples: stock markets, bond markets.
- **Financial security:** document (often electronic) stating terms under which funds pass from buyer to seller.
- **Stock:** financial security representing partial ownership of a firm.
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The Macroeconomics of Saving and Investment

- Start from the national income identity:

$$Y = C + I + G + NX.$$

- For simplicity, assume a **closed economy** (no exports or imports), so:

$$Y = C + I + G.$$

- Rearranging to solve for investment:

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Private Saving and Public Saving

- **Private saving (S_{private}):**

- Households receive income from factors of production (Y) and transfers (TR).
- They pay taxes (T) and consume (C).

- So:

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving (S_{public}):**

- The government receives tax revenue (T),
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$$S_{\text{public}} = T - G - \text{TR}.$$

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Total Saving and Investment

- **Total saving is:**

$$S = S_{\text{private}} + S_{\text{public}}.$$

- Substitute the formulas:

$$\begin{aligned} S &= (Y + \text{TR} - C - T) + (T - G - \text{TR}) \\ &= Y - C - G. \end{aligned}$$

- But from earlier:

$$I = Y - C - G.$$

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Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
 - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
 - Government runs a **budget deficit**.
 - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
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The Market for Loanable Funds

- We can think of all financial markets together as one **market for loanable funds**.
- **Borrowers:** firms that want to invest in capital (and the government when it runs deficits).
- **Lenders:** households that save and lend funds, and sometimes foreign investors.
- The **real interest rate** is the price that equates saving and investment.

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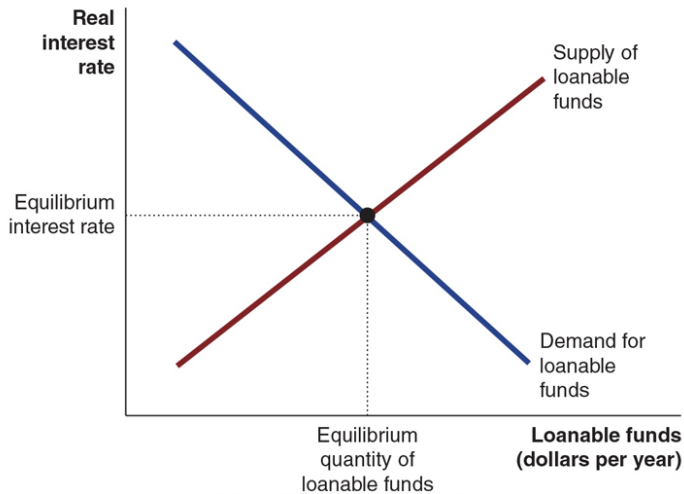
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Figure 6.3 – The Market for Loanable Funds



Supply and Demand for Loanable Funds

- **Demand for loanable funds:**

- Firms borrow more when the real interest rate is lower,
- because more investment projects are profitable.

- **Supply of loanable funds:**

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- Government saving (or dissaving) also affects supply.

Supply and Demand for Loanable Funds

- **Demand for loanable funds:**

- Firms borrow more when the real interest rate is lower,
- because more investment projects are profitable.

- **Supply of loanable funds:**

- Households save more when the real interest rate is higher,
 - because the reward for postponing consumption is greater.
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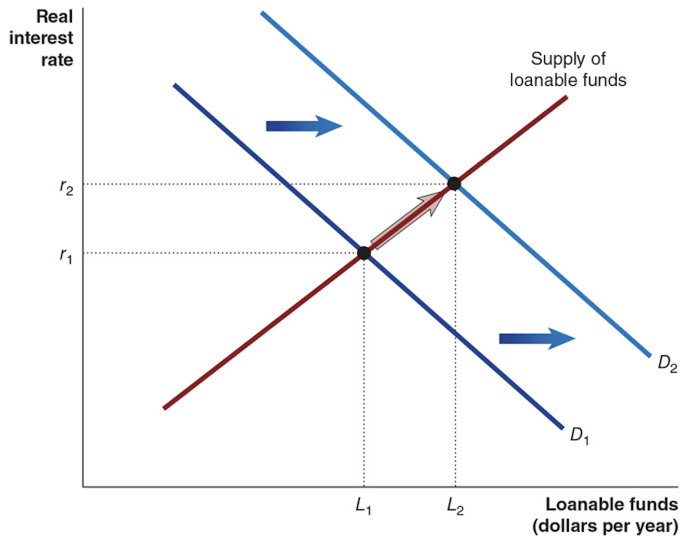
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Figure 6.4 – Increase in Demand for Loanable Funds



Increase in Demand for Loanable Funds

- Suppose technological change makes investment more profitable.
- Firms want to borrow more at each interest rate.
- **Result:**
 - Demand for loanable funds shifts right.
 - Equilibrium real interest rate rises.
 - Equilibrium quantity of loanable funds (investment) rises.

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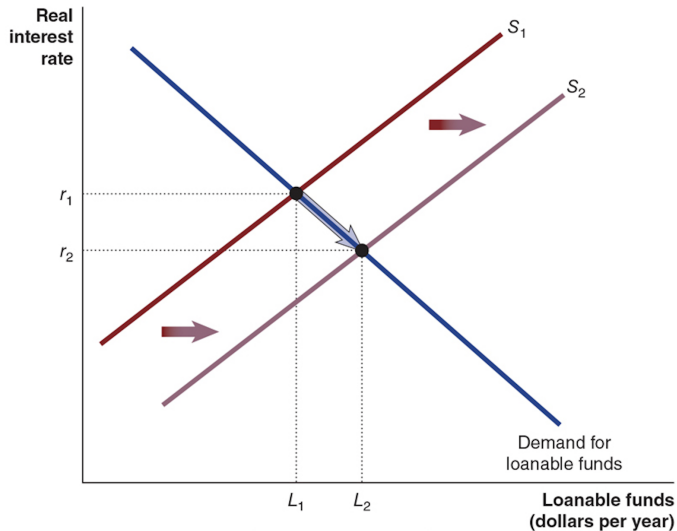
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Figure 6.5 – Increase in Supply of Loanable Funds



Increase in Supply of Loanable Funds

- Suppose households become more patient or more concerned about retirement, and decide to save more at every interest rate.
- Result:
 - Supply of loanable funds shifts right.
 - Equilibrium real interest rate falls.
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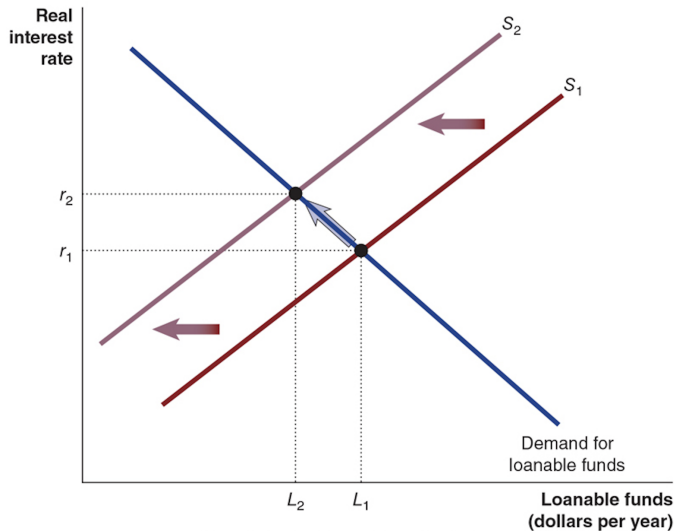
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Figure 6.6 – Effect of a Budget Deficit



Budget Deficits and Crowding Out

- When government runs a **budget deficit**, it must borrow funds.
- This reduces the funds available for private borrowers.
- In the loanable funds market:
 - Supply of loanable funds shifts left.
 - Equilibrium real interest rate rises.
 - Equilibrium quantity of funds to firms falls.
- This reduction in private investment due to government borrowing is **crowding out**.

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How Important Is Crowding Out?

- In practice, the effect of deficits on interest rates is often **small**.
- Why?
 - Interest rates are influenced by **global saving**.
 - Even several billion dollars of extra borrowing may be small relative to the world market.
- Small does not mean zero:
 - Higher public debt may require higher taxes later.
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6.3 Learning Objective

Goal

Explain what happens during the business cycle and how it affects inflation and unemployment.

- Real GDP per capita has risen about threefold since 1961.
- But it did not rise **smoothly**—there were periods of recession and expansion.

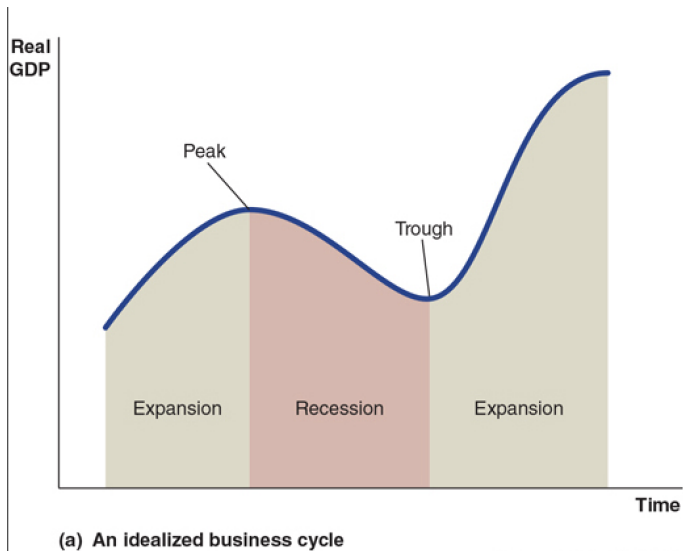
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Figure 6.7 – The Business Cycle (Idealized)



Phases of the Business Cycle

- **Expansion:** real GDP is rising.
- **Peak:** the point where real GDP stops rising and begins to fall.
- **Recession:** real GDP is falling.
- **Trough:** the point where real GDP stops falling and begins to rise.
- After the trough, a new expansion begins.

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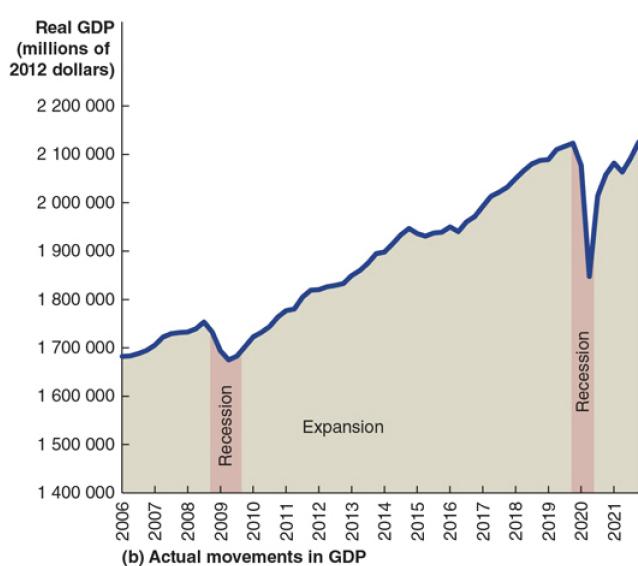
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Figure 6.7 – Canadian Real GDP, 2006–2021



Recent Business Cycles in Canada

- After a peak in 2008:
 - The economy entered a recession from 2008 to 2009.
 - Real GDP fell, and unemployment rose.
- After the next peak in 2019:
 - A particularly deep recession occurred during 2020 due to the COVID-19 pandemic.
 - The trough occurred between April and June 2020.
 - Real GDP only returned to its 2019 level at the end of 2021.

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- Statistics Canada provides many data series (real GDP, employment, income, etc.).
- There is no single Canadian agency that officially declares recessions.
- Common rule of thumb:
 - A recession is two consecutive quarters of negative growth in real GDP.
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The Effect of the Business Cycle on Inflation

- **Inflation rate:** percentage increase in the price level from one year to the next.
- During expansions:
 - Demand for products is high relative to supply.
 - Firms can raise prices more easily $\Rightarrow$ higher inflation.
- During recessions:
 - Demand for products is low relative to supply.
 - Prices increase more slowly or may even fall $\Rightarrow$ low inflation or deflation.

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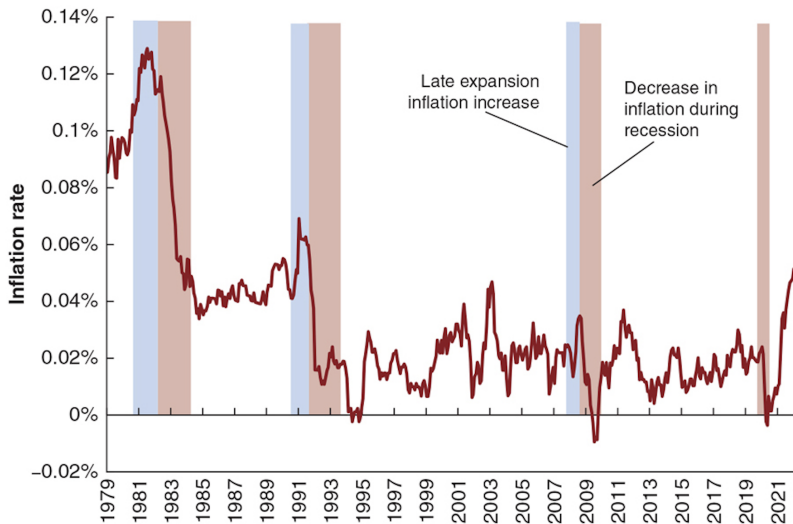
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Figure 6.8 – Recessions and the Inflation Rate



Inflation Over the Business Cycle

- In each of the last three Canadian recessions:
 - Inflation tended to rise during the preceding expansion.
 - Inflation tended to fall over the course of the recession.
- This pattern helps central banks:
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The Effect of the Business Cycle on Unemployment

- During recessions:
 - Firms see sales fall.
 - They reduce production.
 - They lay off workers.
- **Unemployment** often continues to rise even after a recession begins.
- During expansions:
 - Firms hire more workers.
 - Unemployment gradually falls.

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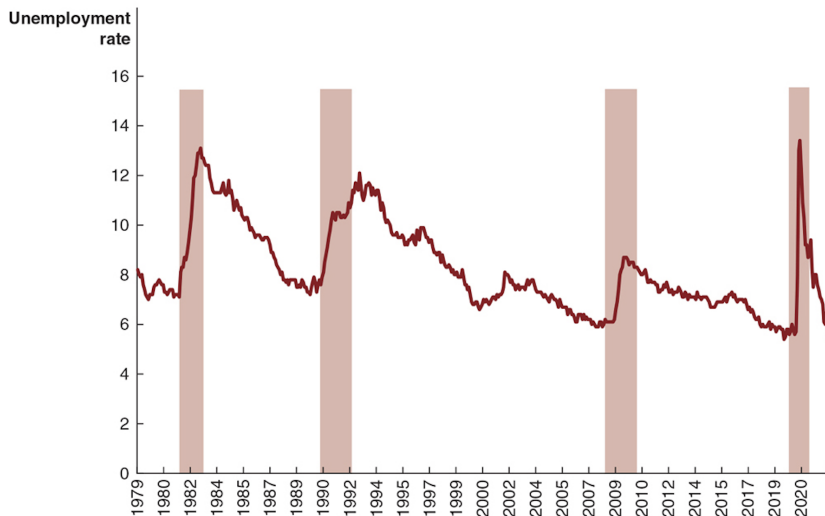
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Figure 6.9 – Recessions and the Unemployment Rate



Putting It All Together

- **Long-run trend:**

- Real GDP per capita grows as productivity rises.

- **Short-run fluctuations:**

- Business cycles around the trend.
- Recessions associated with:
 - lower output,
 - lower inflation,
 - higher unemployment.

- **Financial system:**

- Channels saving into investment, supporting long-run growth.
- Government borrowing can crowd out some private investment.

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Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
 - capital per worker,
 - human capital,
 - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
 - risk-sharing,
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- In a closed economy, **saving equals investment**:
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Thank you!